EXPERIMENT 10

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TITLE:

Create data visualization (pie chart and bar graph) for a student performance system using JavaScript.

AIM:

The aim is to create data visualizations such as pie charts and bar graphs to represent student performance data using JavaScript and Chart.js.

PROCEDURE:

Step 1: Create HTML Structure

Create an HTML file to display charts and include Chart.js library.

HTML CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Student Performance Dashboard</title>
<style>
body {
    font-family: Arial, sans-serif;
    text-align: center;
    background-color: #f4f4f4;
}
canvas {
    margin: 20px;
    background: #fff;
    padding: 10px;
    border-radius: 8px;
}
</style>
</head>
<body>
<h1>Student Performance Dashboard</h1>
<canvas id="pieChart"></canvas>
<canvas id="barChart"></canvas>
<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
<script src="script.js"></script>
</body>
</html>
```

Step 2: Add JavaScript for Visualization

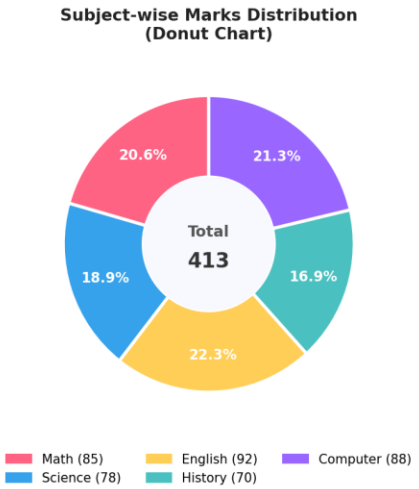
```
const subjects = ['Math', 'Science', 'English', 'History', 'Computer'];
const marks = [85, 78, 92, 70, 88];
const dataset = {
    label: 'Marks Scored',
    data: marks,
    backgroundColor: [
        '#FF6384',
        '#36A2EB',
        '#FFCE56',
        '#4BC0C0',
```

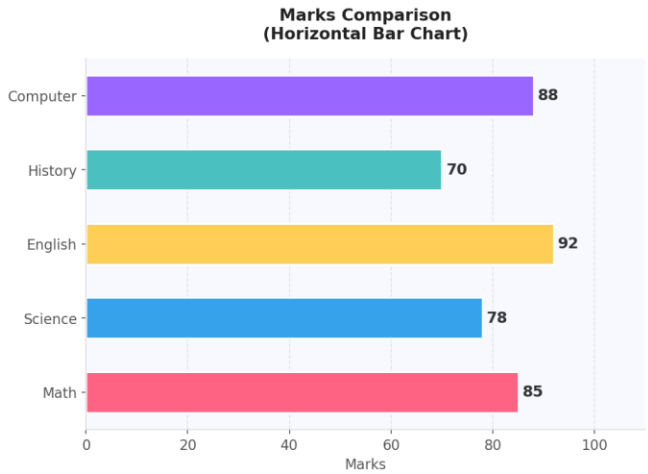
```
        '#9966FF'
    ]
};
// Pie Chart
new Chart(document.getElementById('pieChart'), {
    type: 'pie',
    data: {
        labels: subjects,
        datasets: [dataset]
    },
    options: {
        plugins: {
            title: {
                display: true,
                text: 'Subject-wise Marks Distribution'
            }
        }
    }
});
// Bar Chart
new Chart(document.getElementById('barChart'), {
    type: 'bar',
    data: {
        labels: subjects,
        datasets: [dataset]
    },
    options: {
        plugins: {
            title: {
                display: true,
                text: 'Marks Comparison'
            }
        },
        scales: {
            y: {
                beginAtZero: true
            }
        }
    }
});
```

OUTPUT:

The output displays:

- A donut-style pie chart showing subject-wise marks distribution with percentage labels
- A horizontal bar graph comparing marks across subjects with value labels


Subject-wise Marks Distribution
(Donut Chart)
20.6%
21.3%
Total
413
18.9%
16.9%
22.3%
Math (85)
English (92)
Computer (88)
Science (78)
History (70)


Marks Comparison
(Horizontal Bar Chart)
Computer
88
History
70
English
92
Science
78
Math
85
0
20
40
60
80
100
Marks